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Project Title:	Complex Plasma under Microgravity, Utilizing the International Space Station Experiment “PK-4” and Beyond		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Space Biology Element:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
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COI Name (Institution):	Thomas, Edward Ph.D. (Auburn University) Williams, Jeremiah Ph.D. (Wittenberg University)		
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Task Description:

Complex plasmas are four-component systems that contain electrons, ions, neutral atoms, and charged microparticles (i.e., the dust particles). Depending upon the experimental conditions, the microparticles in these systems can show solid-like, liquid-like, or gas-like properties. However, ground-based experiments are subject to gravitational forces that often dominate the behavior of the systems under investigation leading to sedimentation and anisotropic effects. Therefore, in order to study large-scale complex plasmas and the influences of the weaker forces that are masked by the gravitational force, experiments have to be performed under “microgravity” conditions. Over the past decade, several complex plasma experiments have been performed using the International Space Station (ISS). The latest of these research instruments, “Plasma Kristall 4” (PK-4), is the first microgravity laboratory that is optimized to study fluid phenomena and fundamental effects such as particle charging, screening, and the influence of a directed ion flow. Additionally, since PK-4 is a multi-user facility, we have a unique opportunity to use this instrument to study ion/particle and particle/particle interactions that we otherwise would not be able to address in ground-based experiments.
The intellectual merit of this research project is centered on three major research topics.

Topic 1: Charging and interaction of microparticles in the plasma. The most important measurement in a complex plasma is the measurement of the charge on the particles. However, direct experimental measurement of the charge is difficult, because it usually relies on additional assumptions about experimental parameters such as the plasma electric field and on the validity of an assumed particle interaction type. The microgravity environment of PK-4, where the dust is suspended in the bulk plasma, allows experiments that can potentially lead to a higher accuracy in the particle charge determination.

Topic 2: Ion-dust interactions and the thermal properties of dusty plasmas. A key observation in complex plasmas is the unusually high dust kinetic temperature. It has been postulated that ion-dust interactions are largely responsible for heating the dust particles. The microgravity environment of the PK-4 laboratory, in combination with the tools to actively manipulate the ions in the plasma, provides a unique platform for studying the transfer of energy between ions and the microparticles in the plasma.

Topic 3: Particle transport and heating in waves. The low charge to mass ratio of the dust leads to characteristic frequencies for the dust component that are on the order of a few Hz. This, coupled with the finite size of the dust grains, allows researchers a unique opportunity to directly image plasma physics phenomena and particle transport at the kinetic level. In the microgravity environment of PK-4, it will be possible to measure the transport properties of particles during their initial injection into the bulk plasma, follow the particles as they form collective structures (e.g., waves), and study how ion flow modifies the thermal properties of these waves.

This proposed research project has several opportunities to have a broader impact: (1) through on- and off-campus outreach activities and the training of undergraduate and graduate students, this project will cultivate the next generation of plasma scientists; (2) through the development of new ground-based laboratory instruments (e.g., plasma sources, diagnostics) that will complement the proposed research and contribute to the development of next-generation dusty plasma microgravity experiments; and (3) through the development of new, advanced software tools for image analysis and particle tracking that will support the proposed studies and be offered as community-wide research tools.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research performed under this project falls in the main category of fundamental research. Often, findings obtained by performing fundamental research show their true impact for the society only after many years. However, the study of dusty/complex plasma can benefit the overall research infrastructure in many ways already in the timely vicinity of any discoveries made, mainly as a result of the strong interdisciplinary nature of this research. Studying complex plasma topics such as wave phenomena, heating, the spatial separation of phases, their interaction or the transition between phases such as from fluids to crystals all being analyzed down to the scale of individual particles can lead to fundamental findings that might be immediately applied to many other research fields such as solid state physics, fluid physics, plasma physics, astrophysics. As the specific research performed within the scope of this project makes use of the microgravity (free-fall) environment supported by the national laboratory in space, the International Space Station (ISS), further utilizing an experiment that is build by the European Space Agency (ESA), highly international cooperation activities are a direct result. It is likely that the effort to work on the same fundamental questions in a cooperative acting international team will ultimately benefit each involved nation also beyond just the research. Some aspects of the specific research topics within the scope of this project involve the development of universal software, hardware, or analysis tools that can also be used under different research situations, and some of the more specific technical tools that are developed, such as a plasma impedance probe, might even help to overall improve technology that can be used for industrial purposes. Finally, as part of the general research mission, we train the next generation of engineers and scientists who can in the course of their future career produce a large benefit for the overall economic situation of the nation and beyond.

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Task Progress

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On April 16th 2019 we started our third year of the project. Our team was already well established. Minor changes in the workforce during the year was due to some undergraduate students leaving and some joining our mission. Some participants and some activities are financed from sources outside of the scope of this project. The backbone of this project is its established, fruitful, international collaboration with multiple institutions. Our major partner within this close cooperation is the leading PK-4 core group of the International Space Station (ISS) experiment “Plasma Kristall 4” (PK-4), localized at the German Aerospace Center (DLR) in Wesslingen, Germany. During the reporting year, we even further intensified our collaborative efforts. We joint with Baylor University to define, propose, and execute experiment scenarios for the PK-4 experiment as a team. We shared equipment such as the Baylor PK-4 setup or tested our experiment procedures in a cooperation with the German Aerospace Center (DLR) on their PK-4 ground reference module. The flight setup of PK-4 has been successfully operated now regularly aboard the International Space Station (ISS) since its upgrade in year two of the project. As part of our cooperation with some of the German teams, we were offered to participate on parabolic flight experiment campaigns related to PK-4 by the University of Giessen, Germany, and related to the Ekoplasma experiment, a potential successor for PK-4, by the DLR. Principal Investigator (PI) U. Konopka has been preparing his participation as experimenter/passenger and ground support from the Auburn University side for both experiment campaigns. The campaigns are anticipated to be performed in 2020. Also, the PI intended to join the DLR group early 2020 to perform PK-4 and Ekoplasma related experiments during a two month stay. Unfortunately, this had to be canceled after a few weeks due to the upcoming COVID-19 pandemic.

The progress in our work was as anticipated or in some aspects beyond what we expected. In year three of the project we were able to conduct two experiment campaign aboard the ISS utilizing PK-4 during its campaigns 7 and 9. This was possible because of our streamlined approach with the incorporation of Baylor University to define combined experiments, test them, and execute them together. One campaign was executed in July 2019 and the other one in February 2020. During each campaign multiple experiments, from Baylor, Wittenberg, and Auburn, were combined in a 30 minutes recorded experiment run, resulting in many gigabytes of video and housekeeping data. Due to transmission limitations to the ISS only a minor subset of the experiment data has been available directly after the experiment performance. The hard disks containing the detailed experiment data are usually brought back to ground from the Space Station at a later stage and also have to be distributed to all participants in a time consuming process. For the first experiments we receive the complete data set at the end of 2019. This data is now under analysis. The second data package from campaign 9 in February 2020 has not yet been made available. We expect the data to be available in the summer of 2020.

Beyond experiments aboard the ISS, we also execute experiments using ground based setups similar to PK-4 at Baylor University as well as the PK-4 ground reference module at DLR. Other experiments were performed using the next generation experiment breadboard for Ekoplasma as well as Auburn’s own experiment setups in its “Microgravity Lab.” Additional activities were related to theoretical and some to computation modeling. We for example developed dedicated molecular dynamic simulations for the different research topics.

In the following we give a short summary of the activities/progress details as archived in year three of this project based on it mission to gain new knowledge in the physics of complex (dusty) plasmas and to create a vision beyond the ongoing experiments towards future space experiments on complex plasmas in space (presumably also aboard the ISS). The research that has been conducted is categorized in three major sub topics as detailed in the original proposal:

Topic T1: Charging and interaction of micro-particles in a plasma, Topic T2: Ion-dust interaction and the thermal properties of dusty plasmas, Topic T3: Particle transport and heating in waves, Topic TX: Future space experiments, administrative tasks and analysis tool developments.

We were able to archive the following described progress towards our project goals:

Topic T1: During the third year of the project we continued our efforts towards the measurement of the particle charge following three different approaches that emerged from targeting this goal in the first two years: a) Determining the average particle charge in a dust cloud from the diagnostic of the plasma (measurement of the electron and ion densities) and using the charge neutrality condition to derive the charging state of the dust cloud. b) Analyzing dust-dust collisions within a dense dust cloud and deriving the particle interaction and thus an effective dust charge from first principles. c) Using the Lorenz force in a magnetized plasma to study charge imposed effects on a steady dust flow. Latter resembles an analogue to the classical Hall-Effect.

During year three, Brandon Doyle was able with the PI’s supervision to fully automate the experiment so that an experiment series (pressure or power scan) can be performed in less than a day rather than in a month before. As the pressure or power of the discharge is cycled through a predefined list, the transmission of a signal, scanned over a frequency range from around 1 MHz to 1GHz, through the plasma is recorded via a vector network analyzer. From the details of the plasma response we derive the electron density. As an additional modification to earlier work, we also switched to a radio frequency discharge as this is more common in dusty plasma research. We are now in the process of combining an ion saturation current measurement to the electron density measurement to be able to investigate the charge balance and thus finally the average charge collected by the dust component in a dusty plasma configuration. We expect that the work of Brandon Doyle can be finished in the coming year of the project. Dylan Funk, another graduate student, is studying the effects of Lorenz force on a streaming dusty plasma. He has made some significant progress in 2019/2020. We were able to derive an analytical fluid model that describe the Lorenz force effects on a steady dusty plasma flow. Similar as an electric field is generated in the classical Hall effect across the current direction, the fluid model predicts in the dusty plasma flow the development of a dust density gradient across the flow. The charge of the dust is encoded in this density gradient. To validate the predictions, we did setup a molecular dynamic simulations, which we also further improved to cover a stochastic force. Latter is needed to select a specific coupling parameter, the ration of the average particle potential energy to its thermal energy, for the final steady flow. As the fluid model is only applicable for low coupling values, the fluid phase, we have at the end of year three started to investigating the effect of the coupling on on the observed dust density gradient across the flow direction.

Topics T2/T3: Topics T2 and T3 are performed at a reduced scale compared to the initial proposal. As such we report on the effort on these two topics combined.

In the recent project year we intensified our collaboration effort especially within the US, which has allowed us to streamline our research efforts for PK-4. We succeeded to define, propose, and execute experiment packages for two PK-4 campaigns aboard the ISS. Each of the experiment packages were scheduled their own, exclusively 30 minutes time frame during a PK-4 campaign. One was performed during the 7th PK-4 campaign in July 2019, while the second

was executed in February 2020 during the 9th campaign. The experiment execution was overseen from the experiment team mainly by Lori Scott (AU) and one of the PIs/Co-Is. For the experiment preparation we made use of the ground based experiment setup at Baylor University and we validated the experiment scripts at DLR on their PK-4 ground reference module. We received the first complete data set end of 2019, while the delivery of the data from the 2020 campaign is pending. The data of campaign 7 is now being analyzed.

During campaign 7 we studied the effects of the onset of polarity switching on the velocity distributions in the dust cloud as well as looking for potential produced waves. While the discharge is changed from a direct current to a switched current mode, the particle interaction is altered quasi instantaneously. As the particles have a substantial inertia they cannot follow this fast changes. The system is practically subject to internal strains and stresses in the moment of switching the discharge mode. Energy is potentially introduced into the system, which should manifest in waves and/or in the heating of the system. We started to analyze both effects and their spatial/directional variations using particle image velocimetry techniques. During campaign 9 we utilized the laser manipulation feature of PK-4 to deposit energy in a defined region within the system instead of using the polarity switching technique. The high power laser is switched in a quasi random fashion using a switching frequency, higher than the dust can directly respond to. The expected heating of the dust particle under the influence of this quasi random force will be analyzed again spatially resolved to determine the energy distribution mechanisms in the system. Latter analysis is pending as we wait for the experiment data to be delivered from the ISS to ground.

Meanwhile, we prepare our participation in a parabolic flight campaign performed by Giessen University (group of Dr. M. Thoma) featuring a PK-4 similar setup in an ESA sponsored parabolic flight campaign. This flight originally planned for May 2020 has been postponed due to the COVID-19 pandemic.

Topic TX: In addition to the activities above, we also performed some activities towards the development, definition, and utilization of the next generation space experiment for complex plasma research, "EkoPlasma," formerly named "PlasmaLab." The PI has contributed also in year three to the development of the next generation experiment. The radio frequency generator that was developed in 2019 for the research effort described under topic T1 has been considered to be tested for its applicability for Ekoplasma. Its special capability to control the phase and amplitudes of multiple rf channels would satisfy the requirement of the Ekoplasma project. However, the AU generator momentarily only supports the control of 2 rf-channels while Ekoplasma needs a four channel system. An update to the Auburn development was discussed with the German Aerospace Center. A modified version of the generator might be tested on one of the future parabolic flight campaigns. The PI was offered by DLR to join their next Ekoplasma parabolic flight campaign in 2020 to perform experiments with the setup aboard an airplane during parabolic, free-fall, maneuvers. Preparation for this campaign have started. The rf-generator will not be tested during this campaign though.

Overall: We presented our progress, results, and ideas in the third year of the project on multiple occasions. The two major events that have to be mentioned specifically are the Fundamental Physics Workshop by NASA and the 61st Annual Meeting of the American Physical Society (APS) in the Division of plasma Physics (DPP). During latter meeting we also participate in the connected “Plasma Expo.” As two of our group members are also in the facility science team that overlooks the scientific program and operation of the PK-4 space experiment, we also participate in the annual meeting and the attached scientific gathering with multiple members of our group. The PI of the project further started to explore opportunities to participate in DLR’s and Giessen Universities parabolic flight program related to PK-4 and its successor. Flight are expected to take place in 2020.

We believe that the thirf year was very successful. However, we are somewhat concerned about the impact of the COVID-19 pandemic that manifested itself early 2020. It has already had some impact on our in person cooperation with the German Aerospace Center and Giessen University as the cooperative work (sabbatical) of the PI had to be canceled. Also, at least one of the two participation in parabolic flight campaigns is now postponed. We developed a plan to mitigate the effects of COVID-19 as much as possible, by concentrating our efforts on the theoretical part of our investigations (incl. numerical simulations) as well as on the analysis of the recently obtained data from the space experiment (campaign 7). There is still plenty of work to do that does not depend on potential travel restrictions nor the accessibility of our local laboratories as long as the duration of the impact is limited.

An outlook beyond PK-4: PK-4 will be operational for a couple of more years. With the first two experiment scenarios we initiated a regular workflow to participation in every other campaign. We will continue to do so in the near future and hopefully also beyond the ongoing project, if these activities will further be supported by NASA and the National Science Foundation (NSF). PK-4 should eventually be replaced by a follow-up experiment. Up to recently, a successor experiment, “Plasmalab” alias “Ekoplasma,” was foreseen to be the potential successor. This experiment facility represents the top notch of microgravity dusty plasma research. It is very much different than PK-4 and addresses a variety of different research topics within the field of dusty/complex plasma. It also has the potential to tackle interdisciplinary research questions of boundary research fields such as from astrophysics (dust mitigation, dust contamination prevention on solar system bodies), fluid physics (flow dynamics and structure in the transition from liquids to solids), and solid state physics (crystal structures, phase transitions, defects and their propagation), just to name a few. The development of Ekoplasma was initiated and driven for many years by the Max-Planck Institute of Extraterrestrial Physics, with this project’s PI as initial managing and lead scientist. The development and scientific utilization continued till now by the German Aerospace Center up to a recent industrial study of the project. Unfortunately, this project is now at risk due to lag of funding. We hope that an international effort by the different space agencies will allow to define and fly a reduced version of this experiment, an “Ekoplasma light,” so that the research field of complex/dusty plasma can continue to explore this unusual state of matter in its entirety without the disturbing influences induced by gravity. We are looking forward to an eventful fourth project year.

Bibliography Type:	Description: (Last Updated: 09/30/2025)
Abstracts for Journals and Proceedings	Funk D, Konopka U, Thomas E Jr. "Charge Analysis Techniques for Magnetized Dusty Plasma Flows." Presented at the 61st Annual Meeting of the American Physical Society Division of Plasma Physics, Fort Lauderdale, Florida, October 21-25, 2019. Bulletin of the American Physical Society. 2019;64(11):NP10.00088. http://meetings.aps.org/Meeting/DPP19/Session/NP10.88 , Oct-2019

Abstracts for Journals and Proceedings	Doyle B, Konopka U, Hall T, Thomas E Jr. "Impedance Probe Measurements in Dusty Plasmas." Presented at the 61st Annual Meeting of the American Physical Society Division of Plasma Physics, Fort Lauderdale, Florida, October 21-25, 2019. Bulletin of the American Physical Society. 2019;64(11):NP10.00079. http://meetings.aps.org/Meeting/DPP19/Session/NP10.79 , Oct-2019
Abstracts for Journals and Proceedings	Scott L, Thomas E Jr, Konopka U, Williams J, Pustylnik M, Thomas H. "Investigating the Thermal Properties of a Complex Plasma." Presented at the 61st Annual Meeting of the American Physical Society Division of Plasma Physics, Fort Lauderdale, Florida, October 21-25, 2019. Bulletin of the American Physical Society. 2019;64(11):NP10.00091. http://meetings.aps.org/Meeting/DPP19/Session/NP10.91 , Oct-2019